

NGUYEN HOANG THACH

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Portfolio: [hoangthach.vercel.app](#)

EDUCATION

Ho Chi Minh City University of Technology and Education

Bachelor of Engineering in Information Technology (GPA: 3.5/4.0)

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management, Operating Systems, Computer Networks

Ho Chi Minh City, Vietnam

September 2022 - Present

PROJECTS

Agri-Supplies E-Commerce Platform

[Backend](#) · [Frontend](#)

- Summary:** Architected and developed an end-to-end e-commerce solution for agricultural supplies, combining a high-performance Java Spring Boot backend with a highly scalable React frontend.
- Frontend Architecture:** Engineered a robust **React/TypeScript** UI applying a **Feature-Based** directory structure to ensure maintainability at scale. Implemented a centralized HTTP client using **Axios Interceptors** for secure token management and global error handling.
- State & Data Fetching:** Leveraged **React Query** for efficient server-state synchronization, caching, and optimistic UI updates, ensuring a seamless shopping experience.
- Backend & Design Patterns:** Designed a transaction-safe checkout pipeline (FEFO/FIFO) in **Spring Boot**. Implemented the **State Pattern** for strict order lifecycles and the **Strategy Pattern** for dynamic product variant pricing.
- Security & Real-time:** Built a granular RBAC system with automated permission mapping. Integrated WebSockets/STOMP for instant order status notifications and SePay for automated bank-transfer verification.
- Tech Stack: React, TypeScript, Tailwind, React Query, Java 21, Spring Boot, PostgreSQL, WebSockets.*

E-Learning Platform

[Live Demo](#) · [GitHub](#)

- Summary:** Architected and developed a comprehensive, multi-role e-learning platform, delivering a scalable NestJS backend and an enterprise-level React frontend interface.
- Frontend Architecture:** Engineered a modular **React/TypeScript** application with distinct layout infrastructures (*Admin, Instructor, Student*). Implemented a robust **Role-Based Routing** system (*ProtectedRoute*) to strictly govern access to specialized dashboards.
- Media & UI Components:** Built reusable, intuitive UI components with **Tailwind CSS**, including seamless video/image uploaders directly integrated with Cloudinary for course asset management.
- Backend & Performance:** Designed a modular REST API with **NestJS** and **Prisma ORM**. Integrated **Redis** for distributed caching of course catalogs and enforced API security via Rate Limiting (*Throttler*).
- Payment & Real-time Ops:** Integrated dynamic payment gateways using the Factory Design Pattern and established an event-driven architecture with WebSockets to deliver real-time notifications across all user roles.
- Tech Stack: React, TypeScript, Tailwind, NestJS, Prisma, PostgreSQL, Redis, WebSockets, Cloudinary.*

BlinkyVocab - Gamified Learning Platform

[Live Demo](#) · [GitHub](#)

- Summary:** Developed a full-stack, gamified language learning application, engineering both an interactive React frontend and a highly scalable NestJS backend ecosystem.
- Frontend Architecture:** Built a responsive, type-safe UI using **React, TypeScript**, and Tailwind CSS. Abstracted complex business logic into **Custom Hooks** (*useSpacedRepetition, useGamification*) to maintain clean component design.
- State Management & Data Fetching:** Implemented **React Query** for efficient server-state synchronization and caching, combined with React Context API for global client-state management.
- Backend & Algorithm:** Engineered a personalized learning engine using a hybrid algorithm (Spaced Repetition SM-2 & Forgetting Curve) in **NestJS**, strictly adhering to the Repository Pattern with Prisma.
- Real-time Integration:** Built a centralized WebSocket architecture to seamlessly push instant notifications, live XP streaks, and community feed updates from the backend to the React client.
- Tech Stack: React, TypeScript, Tailwind, React Query, NestJS, PostgreSQL, Redis, WebSockets.*

SKILLS

Programming Languages:	TypeScript/JavaScript, Java, SQL, Python
Frontend Technologies:	React, Tailwind CSS, React Query, HTML5/CSS3
Backend Frameworks:	Spring Boot, Node.js (NestJS, Express.js)
Databases:	PostgreSQL, MySQL, MongoDB, Redis
Tools & Cloud Services:	Git, Docker, REST APIs, WebSocket/STOMP, JWT, Cloudinary
Languages:	English (TOEIC 600)

ACTIVITIES & AWARDS

Winner, 'Best Idea Award' - HCMUTE Hackathon 2025

Spring 2025

- Awarded for developing 'TâmTrò', a full-stack web application designed to provide 24/7 AI-powered mental health support for users.
- Engineered the backend RESTful API using Node.js and Express.js to manage user authentication, real-time chat sessions, and mental health assessment data.
- Implemented the core emotion analysis feature, processing chat history to generate and visualize a user's emotional trends over time.